INCH-POUND
MIL-DTL-16356E
02 July 2013
SUPERSEDING
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18 December 1987

DETAIL SPECIFICATION

GAUGES, DIRECT READING, OBLONG GLASS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers direct reading sight indicators and gauges that use oblong glass panes of the reflex or plain transparent type, including replacement glass and gasket sets.

1.2 Classification. Gauges are of the following types and classes, as specified (see 6.1 and 6.2):

1.2.1 Types. The types of gauges are as follows:

Type A - For continuous vision, plain, double, flat glass.

Type B - For reflex glass, pressures of 300 pounds per square inch gage (psig) and under.

Type C - For non-steam tank level applications or 150 psig or less steam applications where the visible range of a single chamber design is acceptable. Gauges use plain, double flat glass as used in Type A, or reflex glass as used in Type B. Included in Type C are welding pad gauges that are designed to become an integral part of a pressure vessel or tank.

1.2.2 Classes. The classes of gauges are as follows:

Class I - For designed pressure to 700 psig (600 psig nominal) using 1500 psig glass.

Class II - For designed pressure to 1050 psig (900 psig nominal) using 1500 psig glass.

Class IV - For designed pressure to 2550 psig (2500 psig nominal) using 1500 psig glass (intended for nominal 1200 psig service).

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part or identifying number (PIN). For gauges and component gauge parts furnished under this specification, the complete gauges and replaceable component gauge parts are identified by part numbers in accordance with the following and as shown in tables I and II:

M	**16356**	**-**	**X**	**X**	**X**	**XX**	**YY**
Prefix for Military Specification	Specification Number		Type A, B, or C (see 1.2.1)	Class Code (see 1.2.2): 0 = Other than Type A 1 = Class I 2 = Class II 4 = Class IV	Size Code (see table IV)	Part/ Assembly Code (see table I)	Supplemental Codes (see table II) (multiple codes may be used)

Examples: M16356-A1500 = 8⅝-inch glass, Type A, Class I gauge, complete.

M16356-C0723M4 = 11-inch gaskets, Type C, special material.

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TABLE I. Part assembly codes.

Code	Item	Code	Item
00	Gauge, complete	52	Drain valves
11	Glasses	60	Chain sprocket
12	Mica shields	61	Sprocket wheel – 7 inch
21	Glass positioners	62	Sprocket wheel – 8 inch
22	Spacers	63	Sprocket wheel – 9 inch
23	Gaskets (for glass)	64	Sprocket wheel – 10 inch
24	Cover plates	65	Sprocket wheel – 11 inch
25	Gaskets, spiral wound	66	Sprocket wheel – 12 inch
31	Frame body	71	Hand wheel – 7 inch
32	Spring cones or bolt assemblies with captive spring cones	72	Hand wheel – 8 inch
		73	Hand wheel – 9 inch
41	Top connectors	74	Hand wheel – 10 inch
42	Bottom connectors	75	Hand wheel – 11 inch
51	Shutoff valves	76	Hand wheel – 12 inch

TABLE II. Supplemental part assembly codes.

Code	Item
G1	Both pressure and cushion gasket thickness is 1/32 inch
G2	Pressure gasket is 1/32 inch; Cushion gasket is 1/16 inch
G3	Both pressure and cushion gasket thickness is 1/16 inch
M1	Material for both gaskets conforms to MIL-DTL-24696, Type I
M2	Material for pressure gasket conforms to MIL-DTL-24696, Type I; Cushion gasket to MIL-DTL-24696, Type II
M3	Material for both gaskets conforms to MIL-DTL-24696, Type II
M4	Special gaskets (see table V)
NM	Kit does not contain a mica shield
HX	Special cleanliness or detrimental material control requirements apply (see 3.17 and 3.18)
VU	Upper valve
VL	Lower valve
RH	Right hand
LH	Left hand

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

FF-S-85 - Screw, Cap, Slotted and Hexagon Head

FF-S-86 - Screw, Cap, Socket-Head

FF-S-92 - Screw, Machine: Slotted, Cross-Recessed or Hexagon Head

FF-S-200 - Setscrews: Hexagon Socket and Spline Socket, Headless

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and System

MIL-DTL-1222 - Studs, Bolts, Hex Cap Screws, and Nuts

MIL-PRF-6855 - Rubber, Synthetic, Sheets, Strips, Molded or Extruded shapes, General Specification for

MIL-DTL-16232 - Phosphate Coating, Heavy Manganese or Zinc Base

MIL-R-17131 - Rods and Powders, Welding, Surfacing

MIL-DTL-22050 - Gasket and Packing Material, Rubber, for Use with Polar Fluids, Steam, and Air at Moderately High Temperatures

MIL-S-24093 - Steel Forgings, Carbon and Alloy Heat Treated

MIL-P-24503 - Packing Material, Graphitic, Corrugated Ribbon or Textured Tape and Preformed Ring

MIL-P-24583 - Packing Material, Graphitic or Carbon Braided Yarn

MIL-DTL-25988 - Rubber, Fluorosilicone Elastomer, Oil- and Fuel Resistant, O-Rings

MIL-DTL-24696 - Gasket, Sheet, Non-Asbestos

MIL-G-24716 - Gaskets, Metallic-flexible Graphite, Spiral Wound

MIL-DTL-24777 (Classified) - Non-Metallic Materials with Special Requirements

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-22	-	Welded Joint Design
MIL-STD-167-1	-	Mechanical Vibrations of Shipboards Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-767 (Classified)	-	Control of Hardware Cleanliness
MIL-STD-792	-	Identification Marking Requirements for Special Purpose Components
MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil.)

(Copies of MIL-DTL-24777 and MIL-STD-767 may be requested from commandstandards@navy.mil addressed with a need-to-know.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE LOGISTICS AGENCY (DLA) DRAWINGS

02131	-	Bolt Assy, Gage

(Copies of this document are available from DLA Fleet Support, Engineering & Technical Services – Drawings, 700 Robbins Ave. Philadelphia, PA 19111-5092.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-2177525	-	Valve Assemblies 1/4-2 BW, SW900LB
803-5184193	-	Valves 1/4 to 2 IN N.P.S., ANSI
5000-S5100-841604	-	Gage, Boiler Water, 18 Inch Minimum, Dimensional Standard for 600 psi Nominal Pressure and 1200 psi Nominal Pressure

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM. Copies of S0005-AE-PRO-010/EDM are available online at https://nll.ahf.nmci.navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

0989-064-3000	-	Cleanliness Requirements for Nuclear Propulsion Plant Maintenance by Forces Afloat
T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods
S9074-AR-GIB-010/278	-	Requirements For Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of these documents are available online at https://nll.ahf.nmci.navy.mil. This publication can be located by searching the Navy Publications Index for the TMIN without the suffix.)

NAVSUP DRAWINGS

6685-798-5485	-	Shield Assembly

(Copies of this document are available from Navy Inventory Control Point Mechanicsburg 5450 Carlisle Pike Mechanicsburg, PA 17050.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331 - Zero Acceptance Number Sampling Plans

(Copies of this document are available from the American Society for Quality, 600 North Plankinton Ave., Milwaukee, WI 53203 or online at www.asq.org.)

ASME INTERNATIONAL

ASME Boiler and Pressure Vessel Code

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard

ASME B16.34 - Valves Flanged, Threaded, and Welding End

ASME B18.21.1 - Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

(Copies of these documents are available from ASME International, 22 Law Drive, P.O. Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A105/A105M - Standard Specification for Carbon Steel Forgings for Piping Applications

ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High Temperature Service

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High Temperature Service

ASTM A193/A193M - Standard Specification for Alloy Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications

ASTM A194/A194M - Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both

ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High Temperature Service

ASTM A217/A217M - Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure Containing Parts, Suitable for High Temperature Service

ASTM A276 - Standard Specification for Stainless Steel Bars and Shapes

ASTM A307 - Standard Specification for Carbon Steel Bolts and Studs, 60,000 PSI Tensile Strength

ASTM A352/A352M - Standard Specification for Steel Castings, Ferritic and Martensitic, for Pressure Containing Parts, Suitable for Low Temperature Service

ASTM A536 - Standard Specification for Ductile Iron Castings

ASTM B163 - Standard Specification for Seamless Nickel and Nickel Alloy Condenser and Heat-Exchanger Tubes

ASTM B165 - Standard Specification for Nickel-Copper Alloy (UNS N04400) Seamless Pipe and Tube

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ASTM B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM D351	-	Standard Classification for Natural Muscovite Block Mica and Thins Based on Visual Quality
ASTM F104	-	Standard Classification System for Nonmetallic Gasket Materials
ASTM F218	-	Standard Test Method for Measuring Optical Retardation and Analyzing Stress in Glass

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

DIN GERMAN INSTITUTE FOR STANDARDIZATION STANDARDS

DIN 7081	-	Pressure Resistant Oblong Toughened Borosilicate Glass Panes for Sight Glasses for Unrestricted Use at Low Temperature

(Copies of this document are available from DIN German Institute for Standardization, Deutsches Institut für Normung e.V, Am DIN-Platz, Burggrafenstaße 6, 10787 Berlin, Germany or online at www.beuth.de. Foreign sales contact Beuth Verlag.)

SAE INTERNATIONAL

SAE-AMS-2371	-	Quality Assurance Sampling and Testing, Corrosion and Heat-Resistant Steels and Alloys, Wrought Products and Forging Stock
SAE-AMS-2750	-	Pyrometry
SAE-AMS-5520	-	Steel, Corrosion and Heat-Resistant, Sheet, Strip, Foil, and Plate 15Cr - 7.1Ni - 2.5Mo - 1.1Al, Solution Heat Treated, Precipitation Hardenable
SAE-AMS-QQ-S-763	-	Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings

(Copies of these documents are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. Material used in the construction shall be the lightest practicable consistent with the strength required for safety and reliability and shall meet all the requirements of this specification.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 General construction. The gauge and as-assembled fittings, valves, and water connection piping shall be in accordance with the requirements of this specification, and, where applicable, to the ASME Boiler and Pressure Vessel Code requirements for power boiler water level indicators and gauge glass connections and ASME B16.34 for type A valves. The gauge assembly shall indicate variations in liquid level over the full range as specified (see 3.9.3, 3.12.2, and 3.13.1). The gauge shall withstand frequent blowdown and rapid cut-in and shall provide ample strength for the pressure at which the gauge is to operate.

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3.4 Parts. A water gauge shall consist of the following parts:

a. Gauge frames consisting of:
   (1) Body sections containing water and steam chamber with stem, nipples, or tube assemblies.
   (2) Coverplates.
   (3) Glasses, mica shields, gaskets, and glass positioners or spacers, if used.
   (4) Clamping device (spring cone washers, bolts and associated hardware).
b. Top and bottom connectors (if separate from shut-off valves).
c. Top and bottom shut-off valves.
d. Drain valves (Types B and C only).

(NOTE: Drain valves for Type A gauges shall be purchased using 803-2177525 and 803-5184193. Flanged connections provided for drain valves shall be compatible with Navy standard valve flanges.)

3.5 Threaded parts. Minimum commercial standards for threaded parts shall be in accordance with FED-STD-H28.

3.6 Bolts, studs, nuts, and screws. Unless otherwise specified (see 6.2), lockwashers, bolts, studs, nuts, and cap screws shall be in accordance with ASME B18.21.1, ASTM A193/A193M, ASTM A194/A194M, ASTM A307, FF-S-85, FF-S-86, FF-S-92, FF-S-200, or MIL-DTL-1222, as applicable. Any of these units that fasten on the moving parts or that may be subject to the vibration during operation shall be secured by a locking device. Body studs and body stud nuts for valves to be phosphate coated in accordance with MIL-DTL-16232, Type Z, Class 3. Zinc plating per ASTM B633, Type 1, Class 12 (salt spray test not required) is an allowed alternative. Coating/plating is to be performed after machining threads to requirements. Minimum fits shall be UNC2A and 2B in accordance with FED-STD-H28 for Type A. Studs shall be tack-welded to the body or assembly if minimum fits are used. Alternate methods for Class I gauges can be used subject to Navy approval.

3.7 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength. Type B reflex glass is excluded from reversible requirement despite interchangeability. Oblong gaskets for use on the pressure side and coverplate side, that are identical except for thickness and material, may be supplied under Type C gasket requirements (see 3.14.5).

3.8 Welding. Joining of steel parts of the frame bodies or valves by welding shall be required for Type A gauges and optional for Types B and C. Such welding shall be in accordance with S9074-AR-GIB-010/278. Acceptance standards of MIL-STD-2035, Class I shall apply. Weld fittings, where used, shall be in accordance with MIL-STD-22.

3.9 Frame.

3.9.1 Frame body construction. Supplemental blowdown or support water columns shall not be used. When more than one section is used in the frame, they shall not be connected in parallel, but shall have single steam and water connections to the top and bottom fittings, respectively. Sections, if used, shall overlap vertically not more than 2 inches nor less than 1½ inches. The glass of the frame bodies shall be held by a removable cover secured by clamping devices sufficient in number to provide a steam-water tight joint against the pressure for which the gauge is to be used (see 3.12.7 and 3.13.3). The cover shall be removable for replacement of glass sets in the insert recesses. Tolerance of overall size of vision opening in the frame and cover shall be +⅛inch, -0 inch. Width of vision openings shall be not less than ½ inch. Material shall be carbon steel for pressure at least equal to that specified in ASTM A105/A105M or grade WCB of ASTM A216/A216M.

3.9.2 Rigidity. The frame bodies and cover-plates shall be sufficiently rigid to withstand the applicable pressure without dependence on overlapping parts of the cover or on the means of securing the covers to the frame.

3.9.3 Visibility. The lowest point of the visible portion of the gauge shall be at least 1½ inches above the lower edge of the opening in the bottom fitting of the boiler drum, and the highest point of the visible portion shall be at least 1½ inches below the upper edge of the opening in the top fitting to the boiler drum.

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3.9.4 Insert recesses. For Type A and B and Class I and II gauges, recesses for the glasses in the frame bodies and covers shall not exceed the maximum dimensions of the glass (see 3.12.3.1 and 3.13.1.1) by more than 1/32 inch on the width and 1/16 inch on the length. The minimum width and length of the recesses shall be such that a glass having maximum dimensions as specified herein shall be easily inserted. Chambers or frames and covers for all Type A and B gauges shall be such that if a glass of minimum thickness specified in 3.12.3.1 and 3.13.1.1 as applicable is clamped into place in the gauge without gaskets and mica shields, the clearance between the surfaces of the covers and frame body, parallel to the faces of the glass, shall be not less than 1/16 inch. Joint faces for the glasses in frame bodies and covers shall be finished machined surfaces.

3.9.5 Chamber passage. Steam and water passages of the gauge shall have a clear opening of an area not less than that of a 3/8 inch diameter circle. This passage shall produce a pressure drop of less than 3 inches of water under service flow conditions.

3.9.6 Joints. The joints between the glasses and the frame body and between the glasses and the covers shall be fitted with gaskets. Spacers as necessary shall be used to centralize the glass in the body recesses.

3.9.7 Cover-plates (covers). Vertical movement of the cover with respect to the frame shall be limited (such that the weight of the cover does not bear on the glass). The covers shall be stiffened or reinforced to withstand imposed stresses as necessary.

3.10 Gauge fittings and valves.

3.10.1 Nipples and gauge connectors. The frame body or bodies of the gauges shall have a minimum of 3/4 inch outside diameter tubular connecting pieces (nipples, tube assemblies, stems) and shall be made of carbon steel at least equal to or better than that used for the frame. The connecting piece shall connect the frame body assembly to the top and bottom flanges. The length of the connecting tubes shall be made to suit the distance between centers of the top and bottom fittings as installed on the boiler. The required ball-check feature (see 3.10.4) of the bottom fitting shall be incorporated into a bottom gauge connector or fitting.

3.10.2 Top and bottom shut-off valves. Centers of the top and bottom shut-off valves of the gauges shall be in the same vertical centerline. Each top and bottom fitting shall include a shut-off valve to be manipulated by a chain operating a sprocket wheel. Valves shall not back off or drift open at pressures for which rated. Maximum allowable torque to seat valves under all conditions shall be limited as shown in table III.

TABLE III. Maximum allowable torque.

Sprocket Wheel or Hand Wheel Diameter (inches)	Equivalent Torque Tolerance ±5 Foot-Pounds
7	35
8	41
9	48
10	54
11	61
12	68

3.10.2.1 Chain length. A minimum of 40 feet of chain shall be furnished with each Type A gauge used on main propulsion boilers. Chain requirements for other applications shall be as specified (see 6.2). Plug cocks shall not be acceptable as shut off valves for any type gauge. Minimum valve size shall be 3/8 inch. Carbon steel valves in accordance with ASTM A105/A105M, or Grade WCB of ASTM A216/A216M, shall be acceptable for pressure ratings, body, and bonnet.

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3.10.2.2 Seat leakage. Seat leakage shall not exceed 10 cubic centimeters (cm^3) per hour per inch of nominal pipe size (see 4.7.1.3).

3.10.3 Valve closure. Valves shall close by clockwise rotation of the wheel. Hand-wheels for drain valves shall be ductile iron for Type B and C gauges. Sprocket wheels shall be made of the material specified in 3.12.10.e. or 3.13.8.c, as applicable, and shall be sized so that the torque specified in table III shall open or close the valves under all operating conditions.

3.10.4 Automatic ball-check. An automatic ball-check feature shall be included in the bottom fitting of the gauge. This ball-check may be either furnished as part of the bottom shut-off valve or may be enclosed in the bottom gauge connectors or fittings. The ball-check shall be solid ball type with the ball material being Monel™ or equal. Upon breakage of the gauge glass, the ball-check shall close automatically by vertical lift of the ball, such vertical lift not less than $\frac{3}{8}$ inch. The ball shall be not less than $\frac{9}{16}$ inch in diameter and the diameter of the circle of contact with the seat shall not be greater than two-thirds of the diameter of the ball. The ball shall be accessible for inspection. A ball-check feature shall not be included in the top fitting.

3.10.5 Gland packing and piping gaskets. Gland packing for valve stem stuffing boxes shall be in accordance with MIL-P-24503, Type II, Class 2 and MIL-P-24583, Class 2, Grade 1 or N as specified (see 6.2). The packing set shall consist of one graphitic yarn packing ring, followed by two to four graphitic seal rings and one graphitic yarn packing ring. Replacement graphitic seal rings are to be provided with a single 45-degree angle cut. Packing gland nut tightening shall not exceed the following consolidation torque limits: 84 inch-pounds for ½ inch size and 168 inch-pounds for ¾ inch size for Class I and II valves; and 180 inch-pounds for ½ inch size and 240 inch-pounds for ¾ inch size for Class IV valves. Flange bonnet and piping gaskets within the gauge shall be spiral wound gaskets in accordance with MIL-G-24716 suitable for the pressures and temperatures specified herein. Winding material shall be Class A (CRES 304) unless Class B (Inconel) is specified (see 6.2). Where carbon steel glands are used, they shall be zinc plated in accordance with ASTM B633, Type II, SC 4. Salt spray test is not required.

3.10.6 Drain valves. Drain valves of the quantity and size as specified (see 6.2) shall be attached to the bottom fitting in such a manner that the gauge glass shall be drained with the shut-off valves open or closed. Valves shall be in accordance with 803-2177525 for Class 600 carbon steel and 803-5184193 for Class 1500 carbon steel. Parts for valves shall be fully interchangeable with parts shown on the drawing revision in effect at contract award.

3.10.7 Hardfaced surface inspection. Dye penetrant inspection of hardfaced surfaces shall be in accordance with T9074-AS-GIB-010/271 and the acceptance standards of MIL-STD-2035, Class 1 weld.

3.10.8 Yoke bushing locking. A weld bead shall be applied to the top of the bonnet at assembly to act as a stop for the yoke bushing, preventing the bushing from unscrewing.

3.11 Flanges. Mounting flanges shall be in accordance with ASME B16.5. Other flanges used within the gauge assembly shall meet the requirements specified herein.

3.11.1 Flange finishes. Flanges shall be prepared for use with spiral wound gaskets. Flange finishes shall be as follows:

a. Gasket contact surfaces of flanges for use with sheet gaskets, if used, shall have concentric or spiral serration of 63 to 500 Roughness Height Rating (RHR) produced by machining cuts of a minimum of 0.002 inch in depth in the minimum range of 30 serrations per inch of face width. The tips of the serrations shall not exceed 0.010 inch in width.

b. Gasket contact surfaces of the flanges for use with spiral wound gaskets shall have a finish with a circular lay (concentric or spiral) having a roughness not exceeding 500 RHR produced by machining cuts of 0.003 inch maximum depth with not less than 40 cuts per inch of face width.

3.12 Type A.

3.12.1 Configuration. Type A gauges shall consist of frame body-sections, each section having a continuous vision slot viewed through single-piece glass insert sets. Color shall not be used.

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3.12.2 Visibility and dimensional standards. Unless otherwise specified (see 6.2), the gauges shall have a visibility of 18 inches. The gauges shall not use more than two sections for main boiler service.

3.12.3 Glass. Gauge glass shall be heat-resisting, tempered borosilicate, clearly transparent, and shall withstand the minimum test requirements specified in 4.7.7. The glass shall be suitable for service conditions of 600 °F and 1500 psig when protected by a mica shield. Glass shall be plain and shall be for use with steam at 1500 psig for Type A, Class I and II, and Class IV gauges as specified in 1.2. Gauge glass shall be indelibly marked on the side with manufacturers name and "1500 HP STEAM."

3.12.3.1 Shape and dimensions. Unless otherwise specified (see 6.2), 1500 psig glass shall be shaped as shown on figure 1 and shall be 1.313+0.035, -0.007 inches wide, 0.688+0.000, -0.004 inch thick, and of the length, L, in inches as shown in table IV. Metric measurements are for information and identification only. Both ends shall be rounded to a semicircle of a radius equal to one-half of the width. Glass shall have a bevel 1/16±1/32 inch wide by 45 degrees nominal around entire perimeter of both flat faces (see figure 1). To facilitate removal of the glass from its mold, a draft angle of as much as 1 degree is permitted on the sides (see end view on figure 1). In this case, dimensions shall be measured at their maximum width and length.

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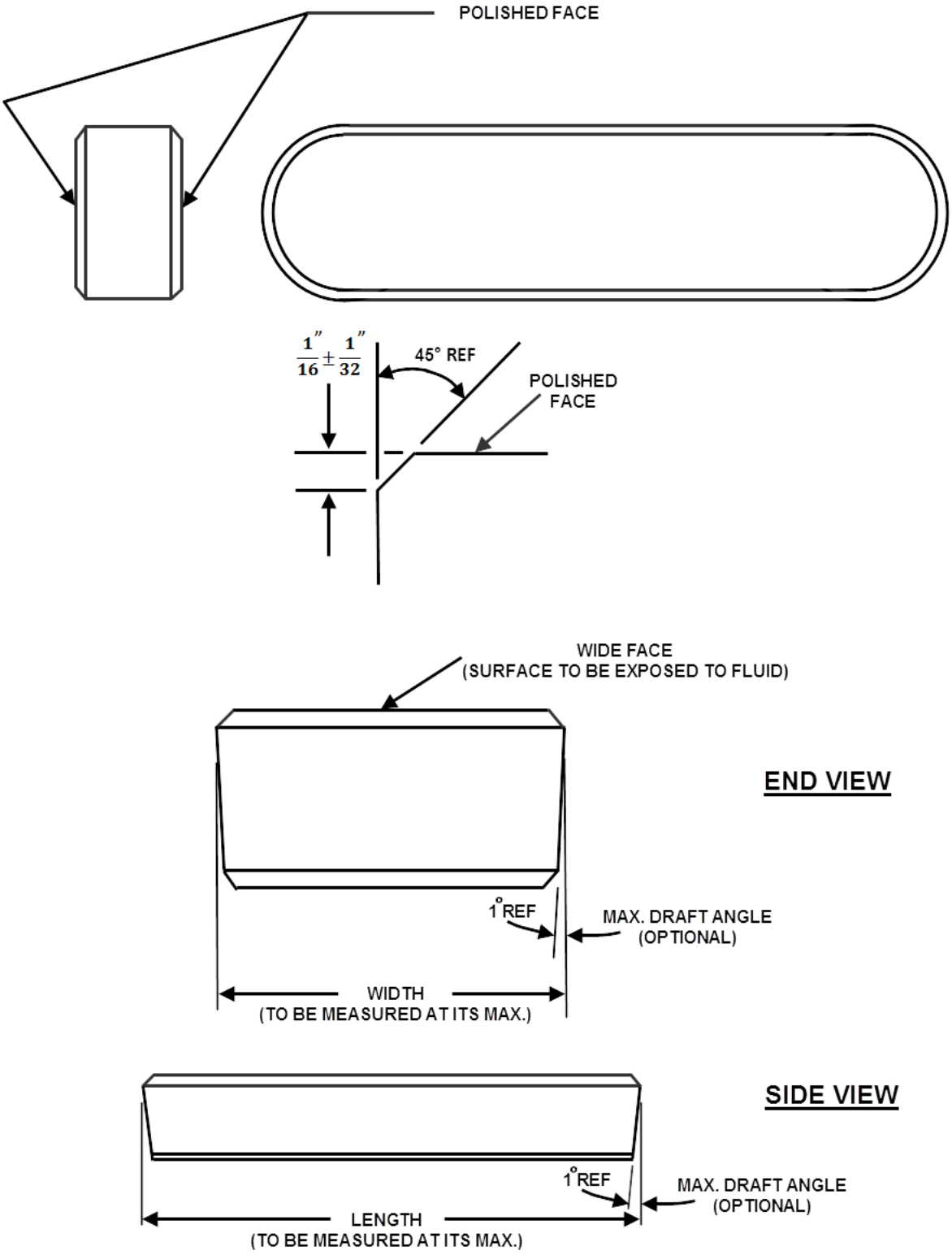


FIGURE 1. Shape and bevel of glasses.

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TABLE IV. Sizes (length) of glasses and type I gaskets.

Size (Number/Code)	L +0.000 -0.031 (Inches)	Metric Size L × W × Thick (Millimeters)
0	3¾	-
1	4½	115 × 34 × 17
2	5½	140 × 34 × 17
3	6½	165 × 34 × 17
4	7½	190 × 34 × 17
5	8⅝	220 × 34 × 17
6	9⅞	250 × 34 × 17
7	11	280 × 34 × 17
7M	11 (22-degree bevel)	280 × 34 × 17
8	12⅝	320 × 34 × 17
9	13⅜	340 × 34 × 17
10	14½	370 × 34 × 17
11	15¾	400 × 34 × 17

3.12.3.1.1 Glass faces. The front and back faces of the glasses shall be parallel within 0.004 inch. All replacement glass supplied under this document shall have a minimum flatness tolerance of 0.002 inch for glass up to and including Size 3, 0.003 inch for glass Sizes 4 and 5, and 0.005 inch for sizes above Size 5. Glass panes meeting these requirements should be acceptable for current Navy applications considered High Pressure (HP) service (see 6.1).

3.12.3.1.2 Glass uniformity. Glass shall contain no scratches or streaks, be uniform in thickness, straight on the sides, free from twist or warp, and shall be free from irregularities which would cause false indication of water level meniscus. The glass shall contain no more than 12 air bubbles (inclusions) over all with a maximum bubble diameter of 0.031 inch, and shall contain no more than six air bubbles within the viewing range of the glass. Air bubbles shall not be located in any pattern as to cause false indication of water level. No more than three solid inclusions are permitted per pane, and they shall not be greater than $\frac{1}{32}$ inch in diameter.

3.12.3.2 Thermal shock resistance. Glass shall not crack or spall when tested as specified in 4.7.7.5.

3.12.3.2.1 Fracture resistance. Glass shall be tempered so that only an interlocking type fracture shall occur when tested as specified in 4.7.7.6.

3.12.3.3 Simulative service. Glass shall be smooth and transparent and shall show no erosion of the surfaces after being subjected to the test specified in 4.7.7.9. Simulative service testing is not required for Class I glass sets using standard reinforced graphite gaskets installed in gauges with conventional flat gasket seating surfaces. Gauges using special design seating surfaces (grooved, tapered, curved, or with less than Ra = 400 finish) will require testing. When simulative service testing is not done, the first article test of 4.7.7.7 and verification test of 4.7.7.8 shall be accomplished on samples of glass from the set.

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3.12.3.4 Special 22-degree bevel glass. This glass, designated Size 7M since it is only used for Size 7 glass on 18-inch visibility main boiler gauges, offers a slightly wider glass surface that has the potential to reduce gasket leakage. When specified (see 6.2), size 7M shall be furnished with a bevel of nominal size $^{1}/_{32}$ inch wide by 22 degrees nominal; a minimum polished flat face width of 1.2 inches; edges of semicircular ends shall be furnished with 22 or 30 degrees (nominal) by $^{1}/_{16}\pm^{1}/_{32}$-inch wide bevel (see figure 2). With the exception of the 45-degree bevel detail, figure 1 requirements also apply to size 7M. This glass is not readily available since it requires unique molds. It should not be bought unless specific Type II and IV gauge requirements exist.

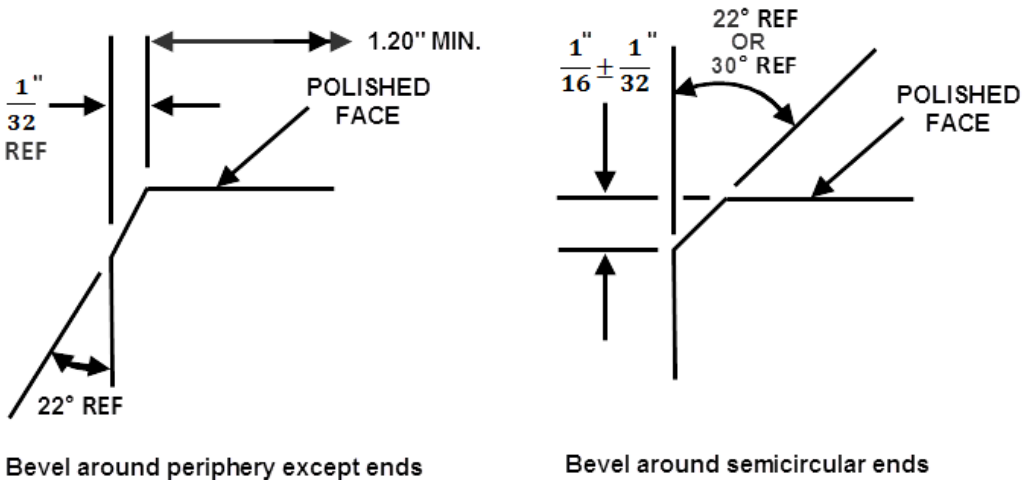


FIGURE 2. Shape of special 22-degree bevel glass.

3.12.4 Mica shield.

3.12.4.1 Mica quality. Mica shall be visual quality classification V-4, good stained, in accordance with ASTM D351 or better, except that air inclusions shall be of V-2 quality or better. The mica shields shall be made from natural muscovite clear white, ruby, light green, or rum-colored mica. Minimum quality of the mica shall be hard, of uniform color and may contain medium waves, slight crystallographic discoloration, and very slight air inclusions in not more than one-fourth of the usable area. Mica shall be completely free from air chains, buckles, ridges, herringbones, cross grains, hairline cracks, tears, sandblast, and holes.

3.12.4.2 Mica shield shape. Unless otherwise specified (see 6.2), a mica shield of the shape shown on figure 1 and of the same width and length as the glass for which intended, shall be furnished for Type A, Class I and II gauges. Mica shields shall be composed of one lamination 0.012+0.002, -0.001-inch thick unless two lamination mica shields meeting 3.12.4.3 requirements are specified. The mica shield shall vary not more than 0.001 inch in total thickness along the length. Class IV thickness requirements are also 0.012 inch. Refer to H6685-798-5485 for other requirements of the Class IV 0.009 +0.003, -0.000-inch thick shield (and gasket) assembly. Two-piece shields are not applicable to Class IV mica and gasket shield assemblies.

3.12.4.3 Two-piece shields. Two lamination mica shields, as a minimum, shall be 0.009 inch thick. The thinner lamination shall be not less than 0.004 inch thick. The combined laminations shall vary not more than 0.001 inch in total thickness along the length. Mica shield packaging shall indicate that the thicker lamination shall be placed next to the steam and water. To facilitate this, the thicker mica lamination shall be marked or identified as such on the packaging. When specified (see 6.2), a two piece mica shield is required.

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3.12.4.4 Mica shield radius. The radius at the ends of the mica shield shall be accurately die-cut and accurately faired into the two parallel sides. Each individual piece of the mica shield shall be free of partial films more than 0.0001 inch thick, and only one partial film shall be permitted in each piece. The complete shield shall have only light visible scratches when hand held at a distance of 6 inches to 1 foot, and only very light visible scratches when viewed at a distance of 6 feet. Edge fractures resulting from cutting or splitting shall be in accordance with the following standards:

Applicable thickness of mica shield piece	Extension of edge fractures from edge of piece
0.004 inch to 0.0075 inch	Not greater than 1/16 inch
Greater than 0.0075 inch	Not greater than 1/8 inch average of lot samples with isolated fractures not greater than 3/16 inch

3.12.5 Gasket.

3.12.5.1 Gasket material. Class I and II steam or water side gaskets shall be made from reinforced flexible graphite material that conforms to MIL-DTL-24696, Type I.

3.12.5.2 Gasket shape. Unless otherwise specified (see 6.2), Class I and II shall be furnished with two identical gaskets 1/32 inch thick that can be used interchangeably as the steam or water side gasket or the outer side or cover gasket, and two spacers 1/32 inch by 5/8 inch by 3 inches made of compressed sheet gasket material conforming to MIL-DTL-24696 Type II. Class I and II gaskets shall be shaped as shown on figure 3. Class IV shall be furnished with two gaskets, one a steam or water side gasket, one an outer side or cover gasket, and spacers as required to conform to gauge frame.

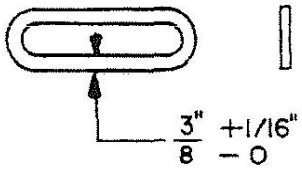


FIGURE 3. Shape of gasket.

3.12.5.3 Outside dimensions and bearing surface of gasket. The outside dimensions of Class I and II gaskets shall be the same as those of the glass for which they are intended. The bearing surface of the gaskets shall be of uniform width and shall be not less than 3/8 inch and not more than 7/16 inch. The bearing surface and dimensions of Class IV gaskets shall be in accordance with H6685-798-5483.

3.12.5.4 Simulative service. When tested as specified in 4.7.7.9, the gaskets shall permit no leakage. Simulative service testing is not required for Class I glass sets using standard reinforced graphite gaskets installed in gauges with conventional flat gasket seating surfaces. Gauges using special design seating surfaces (grooved, tapered, curved, or with more than Ra = 400 finish) will require testing.

3.12.6 Rotation. For purposes of positioning during installation, the gauge frame shall be rotated through 360 degrees.

3.12.7 Frame clamping. Individual means of clamping the body and covers shall be used, compensating for slight buckling or other imperfections throughout the body and cover depth. This shall include individual clamps or fingers, spring cone washers, studs, through bolts, and shouldered cap screws.

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3.12.7.1 Spring cone washers. Spring cone washers shall be used as part of all body and cover clamping designs. Spring cone washers shall be at least equal to 17-7 PH corrosion-resisting steel. Spring cone washers shall be scale free and show no discoloration or lack of uniformity from heat treatment. The final finish shall have no scale, sharp edges, or imperfections. Spring cones as part of shouldered cap screw assemblies for Class IV gauges shall be PH 15-7 Mo alloy in accordance with SAE-AMS-5520. The cap screw assembly with PH15-7 Mo spring cones shall be compression tested in accordance with 4.4.1.8.2. Three randomly selected samples shall be tested for lots up to 150 units. Sample size for lots above 150 units shall be in accordance with ASQ H1331 at Inspection Level S-2. Spring cones used to assemble cap screw assemblies shall be randomly selected. Three sample bolts, assembled with required six washers, shall have stiffness (slope of curve) in the linear region between 0.010 and 0.020 inches of deflection between 110,000 and 225,000 pounds per inch. Variation between the three samples shall not exceed 25,000 pounds per inch. After one deflection to ⅔ of its formed height per 4.4.1.8.2.b, a sample spring cone washer shall return to its original height within ±0.002 inch of its original height. The washers shall withstand the repeated load test specified in 4.4.1.8.2.c without indication of permanent set in excess of 10 percent of the original free height and without cracking or breaking. The heat treatment procedure, to the requirements of SAE-AMS-5520 for austenite conditioning and precipitation heat treatment, shall be tested and approved in accordance with 4.4.1.8.2.a. This testing and approval requires that heat processing equipment meets the pyrometric requirements of a Class 2 parts furnace in accordance with SAE-AMS-2750. The heat treatment facility shall have Nadcap industry certification provided by the Performance Review Institute (www.pri-network.org/nadcap/) in association with SAE International. A furnace Temperature Uniformity Survey (TUS) shall be conducted in accordance with Class 2 requirements described in 4.4.1.8.2.a. Spring cones shall be hardness tested in accordance with 4.7.6.2.

3.12.7.2 Thread protrusion. Gauges fully assembled and tightened for service, with manufacturer's recommended insert materials, shall have all stud or bolt threads protruding beyond the nut by at least two full threads. Stud and bolt lengths shall be established to make thread protrusion equal on all fasteners. Refer to 4.6.1 visual examination requirements.

3.12.8 Chain guides. Shut-off valves on Class I, II, and IV gauges shall be equipped with sprocket wheels and guides. The sprocket wheel shall have a chain guide pivoting from the sprocket center. The guide shall hang below the sprocket in such a way that it will direct the chain on and off the sprocket. Provision shall be made to protect the chain guides and to preclude slippage of the chain from the wheel. In addition for Class IV service, chain stops shall be provided for 3.12.9.2 chain pull requirements.

3.12.9 Gauge fittings.

3.12.9.1 Nipples and connector. The connecting tubes shall be welded into the frame (see 3.9.1). The frame shall be removed from the connectors or valves without removal of the fittings from the boiler to which the gauge is attached. Connectors shall have a minimum of two-hole flanges. The end stem or end connectors shall be slotted so that the frame may be disengaged from the valves or end connectors without springing the expansion loop. The length of the tubular connecting pieces shall suit the distance between centers of the top and bottom fittings as installed on the boiler. Union fittings shall not be permitted. Dimensions shall be in accordance with 5000-S5100-841604.

3.12.9.2 Shut-off valves. One chain pull of 3 feet shall allow sufficient valve opening for spontaneous gauge action. Top valve shall also include expansion loop to allow movement, which is not to exceed ±⅛ inch in the valve frame relationship. Shut-off valves shall use single threads. Nominal size shall be ½ inch. Failure or leakage of the gauge of 200 pounds per hour or less of saturated steam shall cause the bottom automatic check valve to close. Arrangement of the ball-check valves in the gauge assembly shall preclude seating of the balls under pitch and roll conditions up to 60 degrees. The closed bottom check valve shall seal without ball extrusion, seat deformation, or leakage of more than 100 centimeters cubed per hour.

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3.12.9.2.1 Bonnet design. Valves shall be of flanged bonnet construction or bonnetless design. Flanged bonnet construction shall incorporate the following features:

a. Bonnet alignment shall be obtained by body guiding, that is, a close tolerance fit between the bonnet and valve body. The bonnet shall not be aligned by means of the bonnet studs.

b. Flanged face shall assure proper compression of the flanged gasket when the body and bonnet flanges are drawn up metal-to-metal.

3.12.9.2.2 Stem and disc design. Outside screw and yoke construction shall be used. The disc shall be attached to the stem with a swivel. The stem shall have specific hardness depending on material (see 4.7.5). Valves shall have a positive backseat. See 4.7.1.3 for backseat leakage test.

3.12.9.2.3 Stuffing box design. Stuffing boxes shall be of ample depth for at least six turns of packing. Stuffing box glands shall be secured and adjusted by a bolting arrangement which by adjustment shall ensure tightness of the stuffing box under all operating conditions. The bearing surface between the gland follower and gland flange shall be spherically finished.

3.12.9.2.4 Seat and disc design. Seats shall be integral with the body or shall consist of a tubular insert, or seat, welded or brazed into the body. Class I and II valves may use seats screwed into the body. Requirements and alternate materials shall be in accordance with valve material requirements for discs of 803-2177525 apply. Disc and seat shall be hardfaced with Stellite® no. 6, or equal, (in accordance with MIL-R-17131, Type MIL-RCoCr-A-1, -2, and -3), Haynes® alloy no. 25, or equal. Finished thickness of facing shall be not less than 3/32 inch. Solid Stellite®, or equal, disc fabrication is allowed.

3.12.9.3 Drain valves. Two Navy standard valves shall be provided for attachment to the bottom shut-off valve as drain valves, allowing the gauge glass to be drained with the shut-off valves open or closed. The valve closest to the gauge shall be a throttling valve type; since it is adjusted to reduce blowdown. The valve connecting to the piping shall be of the stop valve type since it is used to accomplish periodic blowdown. At least one connection on valve attaching to the gauge shall be flanged; interconnection of the two valves need not be flanged.

3.12.9.4 Mounting flanges. Flanges of top and bottom shut-off valves shall be ¾ inch size. Unless otherwise specified (see 6.2), flange pressure rating shall be 600 psig for Type A, Class I and II gauges and 1500 psig for Type A, Class IV. Drain valve flanges shall also be ¾ inch size.

3.12.10 Materials. Minimum material requirements for Type A shall be as follows:

a. Frame bodies and connector fittings of forged steel, cast steel, steel plate, or pipe shall be of a quality to meet the performance requirements of this specification. Cover-plates shall be forged steel. Preferred material for Class IV gauges is corrosion-resisting steel or carbon steel body with gasket seating surface inlaid with corrosion-resisting steel. Gasket-contact surfaces of carbon-steel bodies, if used, shall be inlayed by a method that results in crack-free bodies.

b. Bolts, studs, and nuts for frame covers shall be steel in accordance with MIL-DTL-1222 or ASTM A193/A193M and ASTM A194/A194M; cap screws shall be in accordance with FF-S-86.

c. Ball-check bodies shall be corrosion-resisting steel in accordance with ASTM A217/A217M or ASTM A276; Haynes Stellite® no. 3; or equal.

d. Tubular connecting pieces shall be material equal to or better than that used in frame body.

e. Valve sprocket wheels shall be ductile iron, forged steel, or cast steel.

f. Valve discs shall be MIL-S-24093, Type V, Class H; SAE-AMS-QQ-S-763, Type 410; Carbon steel ASTM A105/A105M with hardfacing; or solid Stellite® No. 6 or equal.

g. Valve stems shall be ASTM A182/A182M GR F6A modified or SAE-AMS-QQ-S-763, Class 410, Condition T.

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3.12.11 Illuminators and vision hoods. Gauges shall be furnished complete with illuminators operating on 110 volts alternating current (AC). Illuminator boxes and vision hoods (if used) shall be easy to remove and to maintain and depth of box shall be held to the minimum possible. Mercury vapor type lamps shall not be permitted. Illuminators and hoods shall not be adversely affected by externally imposed vibrations through 33 hertz (Hz). Illuminators and hoods shall be shockproof (see 4.4.1.1 and 4.4.1.2). Hoods, when used, shall allow a minimum angle of 90 degrees visibility. Where LED light units are supplied, they must be mounted so that adequate cooling is provided and quick replacement can be achieved in the event of a unit failure. Where two units are provided for 18-inch visibility gauge they shall be interchangeable; they should also be interchangeable with the spares provided. Since there is no illuminator housing on LED illuminators for the installation of emergency back-up lighting, an uninterruptable power supply (UPS) shall be used with LED systems.

3.12.12 Shock and vibration.

3.12.12.1 Shock. Gauges shall withstand the test for Grade A and Type A equipment as specified in MIL-S-901. Gauges shall not be damaged or caused to malfunction by the shock loadings encountered aboard ship, such as those resulting from non-contact underwater explosion, when tested as specified in 4.4.1.1.

3.12.12.2 Vibration. Gauges shall withstand the Type I vibration test in accordance with MIL-STD-167-1 through 33 Hertz (Hz). Gauges shall not be adversely affected by externally imposed vibrations when tested as specified in 4.4.1.2.

3.12.13 Pressure.

3.12.13.1 Class I. Class I gauges shall accommodate a designed pressure to 750 psig (600 psig nominal). These gauges shall be for installation on boilers with up to and including 750 psig design pressure.

3.12.13.2 Class II. Class II gauges shall be for designed pressure to 1050 psig (900 psig nominal). These gauges shall be for installation on boilers with between 700 and 1050 psig design pressure.

3.12.13.3 Class IV. Class IV gauges shall be for designed pressure to 2550 psig (2500 psig nominal). These gauges shall be for installation on 1200 psig boilers and catapult accumulators.

3.12.14 Life. Boiler-water gauge assemblies, except for glass set insets, shall have a life of 40,000 hours of operation at saturated steam pressures corresponding to applicable boiler design pressure. Glass set inserts shall have a minimum life of 5,000 hours of operation at saturated steam pressures corresponding to applicable boiler design pressure. Bulbs for illuminators shall operate for a minimum of 2,500 hours of service. LED illuminator assemblies shall be designed for minimum life of 10,000 hours of service.

3.13 Type B.

3.13.1 Configuration. The Type B gauges shall be furnished with reflex glasses. Sizes and visibility range of gauge glass shall be as specified (see 6.2). A shut-off valve with lever pull and chain shall be furnished for main boiler and catapult accumulator applications (see 3.10.2).

3.13.1.1 Reflex glass. The reflex glass shall be made of a tempered borosilicate glass for use with steam up to 300 psig pressure. The steam or water face of the glass shall have molded or cast prismatic serration extending the full length of the inner or exposed surface of glass. There shall be ample margin on the inner surface to receive the gasket. The glass shall be $1\frac{5}{16}+\frac{3}{64}, -\frac{1}{64}$ inches wide and $\frac{11}{16}+0, -\frac{1}{32}$ inch thick. Length shall be in accordance with size shown in table IV as specified. Both ends shall be rounded to a semicircle of a radius equal to one half of the width. The front and back faces of the glass shall be parallel within 0.010 inch.

3.13.1.2 Gaskets. Two gaskets $\frac{1}{16}$ inch thick made of MIL-DTL-24696, Type I reinforced graphite gasket material shall be furnished with each glass. The outside dimensions of the gaskets shall be the same as those of the glass for which they are intended. The bearing surface of the gasket shall have a uniform width of not more than $\frac{3}{8}$ inch and not less than $\frac{5}{16}$ inch.

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3.13.2 Frame. The frame of the gauges shall have ¾ inch outside diameter tubular connecting pieces. The connecting pieces or tube assembly shall be inserted into the top and bottom valve connections through union fittings. The gauge shall be removed from the valves without removal of the fittings from the boiler. Commercial standard threaded union fittings shall be permitted.

3.13.3 Clamping device. The frame shall be clamped to the cover by U-bolts, the U-bolts extending through the cover and retained by nuts. Means shall be included in the frame or cover to retain the U-bolts in position when the cover is removed. Type A hardware requirements of 3.6 shall apply.

3.13.4 Valves. Type B gauge valves shall be constructed for 300 psig. The quantity, type, and size of drain valves shall be as specified (see 6.2).

3.13.5 Mounting flanges. Flange pressure rating for Type B shall be 300 psig.

3.13.6 Shock. Gauges shall be in accordance with the Type C shock requirement of MIL-S-901 (see 4.4.1.1). When specified (see 6.2), shock testing is required.

3.13.7 Vibration. Gauges shall be in accordance with the Type I vibration requirements of MIL-STD-167-1 through 25 Hertz (Hz) (see 4.4.1.2). When specified (see 6.2), vibration testing is required.

3.13.8 Materials. Materials for Type B gauges shall be as follows:

a. Frames, covers, tube assemblies, and fittings:
   (1) Ductile iron in accordance with grade 60-45-10 of ASTM A536.
   (2) Forged steel in accordance with ASTM A105/A105M.
   (3) Cast steel in accordance with Grade WCB of ASTM A216/A216M or ASTM A352/A352M.
   (4) Steel pipe in accordance with ASTM A106/A106M.
b. Tubular connecting pieces:
   (1) Brass.
   (2) Corrosion-resisting in accordance with ASTM A276.
   (3) Nickel-copper-alloy in accordance with ASTM B163 or ASTM B165.
   (4) Steel pipe in accordance with ASTM A106/A106M.
c. Valve sprocket wheel shall be made of ductile iron, cast steel, or forged steel.
d. Quantity, type, and size of drain valves shall be as specified (see 6.2).

3.14 Type C. Intended for non-steam tank level applications or 150 psig or less steam applications where the visible range of a single chamber design is acceptable. Continuous solid bar chambers with multiple vision slots shall not be provided for boiler applications. Gauges shall use continuous vision, plain, double flat glass, as used in Type A or reflex glass as used in Type B. Included in Type C are welding pad gauges that are designed to become an integral part of a pressure vessel of tank.

3.14.1 Configuration. Type C gauges of the welding pad type shall be furnished with Type B reflex glass or a single transparent pane when installation is made on both sides of the tank to permit see-through visibility. Type C gauges of one piece chamber designs can be supplied with either reflex or double transparent glass as required for the design.

3.14.2 Frame. The connecting pieces or tube assembly shall be inserted into the top and bottom valve connections through union fittings. Tapered pipe threads (NPT) shall not be used anywhere on the gauge or valve connections. Gauge shall be removable from the valves without removal of the fittings from the tank or boiler. Bar stock pad of welding pad gauges shall be of the same base metal as the tank or pressure vessel to which it is to be welded in accordance with grouping of S9074-AR-GIB-010/278. When specified (see 6.2), an insert bar is to be provided.

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3.14.3 Clamping device. The frame of reflex gauges (except welding pad design) shall be clamped to the cover by U-bolts, the U-bolts extending through the cover and retained by nuts. Means shall be included in the frame or cover to retain the U-bolts in position when the cover is removed. Bolting used in all Type C designs shall be in accordance with 3.12.10.b requirements.

3.14.4 Glass. Refer to requirements of 3.12.3 or 3.13.1.1 as applicable for glass requirements. Replacement transparent glass shall be suitable for HP service regardless of pressure rating of Type C indicator.

3.14.5 Gaskets. Two gaskets of 1/16 or 1/32 inch thickness shall be furnished with each glass unless otherwise specified (see 6.2). The outside dimensions of the gaskets shall be the same as those of the glass for which they are intended. The bearing surface of the gasket shall have a uniform width of not more than 3/8 inch and not less than 5/16 inch. Type C indicators may use a variety of gaskets based on the fluid service, in accordance with Table V. In addition to those listed below, the MIL-DTL-24696, Type I and II gaskets supplied with Type A and B gauges may be selected. Type II gaskets can be used in steam service up to 150 psig per 366 °F, water service up to 400 psig per 300 °F, lube oil system up to 150 psig per 250 °F and hydraulic or other petroleum based oil systems up to 600 psig per 180 °F. Where detrimental material requirements of MIL-DTL-24777 apply, special test requirements in addition to those listed in the line callouts are required.

TABLE V. Alternative gasket material.

Commercial Name/ MIL-SPEC (Cage Code)	Thickness (Inch)	Description - Binder	ASTM F104 Line Callout
Thermo-Tork® TN-9004 (Interface Solutions Inc., Lancaster, PA) or equal	1/16	Non-asbestos fiber-composite elastomer - aramid - Nitrile Butadiene Rubber (NBR)	F729100E33M9
IFG®-5500 (73680) or equal	1/16	Non-asbestos fiber - composite elastomer - aramid - NBR	F712103A9B4E23L501M4
Gylon® 3500 - Fawn (73680) or equal	1/16	Polytetrafluoroethylene (PTFE) with silica	F452111A9B4E11M5
Gylon® 3504 - Blue (73680) or equal	1/32 or 1/16	PTFE with glass micro spheres	F456111A9B7E11M6
MIL-R-3533 Type II, Grade A	1/16	Buna-N	N/A
MIL-DTL-25988 Type I, Grade 70/1	1/16	Fluorocarbon	N/A
MIL-DTL-22050 Grade 1	1/16	Ethylene Propylene Rubber (EPR)	N/A
MIL-PRF-6855F Class 1, Grade 60	1/16	NBR	N/A

3.14.6 Material. Material that does not conform to an ASTM grade shall not be used in pressure containing, high stress, or cyclic stress applications. Type A requirements of 3.6 apply to hardware.

3.14.7 Valves. Type C valves shall be rated for either 150 psig or 300 psig to match the rating of the boiler or tank flanges. Drain valves that are part of the welded assembly formed with the lower gauge isolation valve shall be bolted bonnet-type. Gauges for waste heat boilers shall have bolted bonnet shut-off valves. The quantity, type, and size of drain valves shall be as specified (see 6.2).

3.14.8 Shock. When specified, gauges shall be in accordance with the Type C shock requirements of MIL-S-901 (see 4.4.1.1 and 6.2).

3.14.9 Vibration. When specified, gauges shall be in accordance with the Type I vibration requirement of MIL-STD-167-1 through 25 Hertz (Hz) (see 4.4.1.2 and 6.2).

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3.15 Designation and marking.

3.15.1 Water level marker. A marker shall be furnished with each gauge labeled "Normal Water Level." The marker shall be complete with simple means of mounting on gauge frame. Markers shall not be required with replacement frames. Level indicating plates, numbered in inches (plus + and minus -) to identify water level below and above the "N" (Normal Water Level) marker, shall be supplied when specified (see 6.2). There shall be one plate for each gauge segment.

3.15.2 Identification plates. Identification plates and other designating markings shall be furnished. Plates shall be installed on and furnished as part of the boiler-water gauge for which they are intended. They shall be attached to a part of the water gauge which shall not ordinarily be removed during the normal service life, and shall be located in a readily accessible position where they can be read at all times without danger to personnel. Data marked on identification plates shall include the following:

a. Manufacturer's name, design rating, National Stock Number (NSN), manufacturer's contract number, and date of manufacture.

b. Blank space for inspector's official stamping.

c. Specification of shut-off valve as Upper or Lower.

d. Specification of shut-off valve as Right Hand (RH) or Left hand (LH).

3.15.3 Bolt head marking (type A, class IV gauges). Bolt assemblies with captive spring cone washers shall be marked with the manufacturer's symbol and lot number on the top surface of the bolt head. The lot number marking shall define the heat treatment lot of the spring cone washers as well as the lot of the bolt assembly. Markings shall be permanent, as defined in MIL-STD-792, and clearly visible on the finished bolt.

3.16 Tools. For Type A gauges, a kit of tools shall be furnished for each fireroom of the ship for which the gauges are furnished. This kit shall be furnished in a pocketed roll-up kit or in a metal box. One each of all special tools required for assembly or for normal maintenance, repair, and adjustment of the gauges shall be furnished (see 6.10.1).

3.17 Cleaning. Material shall be cleaned so that all surfaces shall be free of corrosion oil, grease, and all other foreign residues. Cleaning shall be accomplished in any manner that will accomplish thorough cleaning without damage to the gauges or gauge parts. Surfaces shall be examined visually to determine freedom from dirt, corrosion, oil, grease, or foreign residue. Preservatives shall not be used on parts that are vulnerable to damage by contact preservatives. When specified (see 6.2), a tag shall accompany each item with the following note: "Prior to installation, this item shall be cleaned in accordance with MIL-STD-767 or 0989-064-3000 to suit the ship system cleanliness requirement." Cleaning in accordance with MIL-STD-767 or 0989-064-3000 is not required unless specified (see 6.2). Spring cone washers shall be vapor degreased or solvent cleaned, followed by mechanical scrubbing with mild abrasive cleaners prior to heat treatment. All traces of cleaners shall be removed by rinsing thoroughly with warm water. A light, tightly adherent, uniform appearing oxide after heat treatment is evidence of proper cleaning.

3.18 Detrimental material requirements. Material supplied under this document shall not contain, and shall not be contaminated by, mercury or mercury compounds. The detrimental material requirements of MIL-DTL-24696 apply to gasket and spacer materials supplied under this document. When specified (see 6.2), packing material in accordance with 3.10.5 shall be supplied Grade N (nuclear).

3.19 Toxicity. When evaluated in accordance with 4.8 (the HHA), the material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.8 and 6.2).

3.20 Workmanship. Gauges shall be uniform, clean, smooth, free of grooves, indentations, cracks or blisters, or any imperfection that might impair serviceability.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of the examinations and tests as specified in 4.4 and 4.6.1.

4.2.1 Sample for first article inspection. A sample gauge shall be examined and tested as specified in 4.2 (see 6.5). Replacement glass sets for Type A shall be examined and tested as specified in 4.7.7.

4.2.1.1 Type A gauge. One unit of each class shall be tested at laboratories having a boiler test stand, an operating boiler, and shock and vibration testing equipment. Laboratory tests shall include tests specified in 4.4.1.1 through 4.4.1.7.

4.2.1.2 Type B gauge. One Type B gauge shall be subjected to first article inspection. First article inspection shall consist of the tests and examinations specified in 4.6.1 and 4.7.

4.2.1.3 Type C gauge. One Type C gauge shall be subjected to first article inspection. First article inspection shall consist of the tests and examinations specified in 4.6.1 and 4.7. Boiler drum diameter shall be suitable to test requirements of 3.12.2.

4.2.1.4 Gauge testing. In addition to any test stand testing performed by the laboratory, the Type A gauge shall be tested by installation on a boiler of the applicable pressure rating for the class being tested. Testing shall include the following:

a. Ability of shut-off valves to close within the allowable leakage (see 4.4.1.3 through 4.4.1.5).

b. Ability of ball-check to seat automatically within leakage specified (see 4.4.1.6).

c. Effectiveness of gauge in indicating boiler-drum water level (see 3.3).

d. Gauge operation on operating boiler (see 4.4.1.7).

e. Shock and vibration testing of gauge, valves, illuminators, and hood, if used (see 4.4.1.1 and 4.4.1.2).

4.2.1.5 Type A glass sets. First article inspection of Type A glass sets shall consist of the tests as specified in 4.2.1.1 and 4.7.7.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.6 and the tests of 4.7.

4.4 First article tests.

4.4.1 Type A gauge.

4.4.1.1 Shock. Type A gauge, complete with valves, illuminators, and vision hoods, if used, shall be subjected to Grade A, Type A tests in accordance with MIL-S-901 (see 3.12.12.1). The assembly shall be pressurized and supported in the same manner as it shall be in the service. Unless otherwise specified (see 6.2), shock test shall not be required for Type B and C gauges.

4.4.1.2 Vibration. Type A gauges complete with valves, illuminators, and vision hoods, if used, shall be tested though 33 Hertz (Hz) in accordance with Type I of MIL-STD-167-1 (see 3.12.12.2). The equipment shall be pressurized and supported in the same manner as it shall be in service. Unless otherwise specified (see 6.2), vibration test shall not be required for Type B and C gauges.

4.4.1.3 Shut-off and drain valve operation. The valves shall be opened and closed 100 times with saturated steam at design pressure on the inlet side and with the discharge connection arranged to permit flow under conditions simulating the maximum flows through the gauge assembly. The opening and closing shall be made with the valve hot and in cycles; that is, starting with the valve open and thoroughly warmed, it shall be closed with a recorded torque, observed for tightness, then re-opened with a recorded torque and the procedure repeated. Packing shall be tight enough to prevent leakage.

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4.4.1.4 Shut-off and drain valve securing. Five securing trials shall be made to determine if the valve remains tight during and after cooling, having closed tight with 15 foot-pounds of torque while hot. Saturated steam at design pressure shall be applied to the inlet side of the valve. The necessity for taking up on the valve stem to maintain specified seat tightness shall be noted. More than a 5 foot-pound increase in torque required to maintain seat tightness and to re-open shall be cause for rejection.

4.4.1.5 Shut-off and drain valve warming. Five warming trials shall be made to determine the torque required to open the valve after it has been closed tightly when cold.

4.4.1.6 Ball-check steam operation. Five securing trials shall be made to determine if the valve remains tight during and after cooling, having closed tight with 15 foot-pounds torque while hot. Saturated steam at design pressure shall be applied to the inlet side of the valve. The necessity for taking up on the valve stem to maintain specified seat tightness shall be noted. More than a 5 foot-pound increase in torque required to maintain seat tightness and to re-open shall be cause for rejection.

4.4.1.7 Assembled gauge operation (type A). Minimum requirements of operation test shall be as follows. Minimum operation of the gauge shall be 1000 hours:

a. Warm gauge at 250 °F per hour to design temperature and pressure with water level maintained at the mid-point of the visible range.

b. Every 8 to 12 hours cool gauge to ambient temperature at rate of 250 °F per hour to design temperature and pressure.

c. Every 50 cycles drop pressure from design pressure to 300 psig in 30 seconds maximum and return to design pressure.

d. Every 20 operation hours maximum while at design pressure, blow down the gauge assembly through the drain valves.

e. Every 200 hours, cool to ambient, isolate and drain gauge, and allow to dry for 48 hours.

f. Repeat steps a through e four additional times.

g. Any additional operational procedure deemed necessary shall be demonstrated once in each 200-hour period.

h. Boiler and test stand testing shall include 50 warm-ups and cool-downs (see 4.4.1.7.a and 4.4.1.7.b), 50 pressure changes (see 4.4.1.7.c), and 50 blowdowns (see 4.4.1.7.d).

4.4.1.8 Type A, class IV gauge, cap screw (bolt) assembly with captive spring cone washers.

4.4.1.8.1 Heat treatment procedure testing. The Temperature Uniformity Survey (TUS) shall use a load that consists of the FAT spring cone washer lot. In accordance with SAE-AMS 2750 load condition requirements, sensors may be attached to simulated parts equal to the spring cone thickness of 0.126 to 0.131 inch. A spring cone washer adjacent to each sensor shall be marked for hardness testing. An equal number of spring cones located midway between sensors shall also be uniquely marked. After heat treatment the marked samples shall be hardness tested. Two Rockwell C hardness tests shall be done on each washer 180 degrees apart on the concave face. At the vendor's option, standard ½-inch wide tensile specimens in accordance with SAE-AMS-2371, cut from the bulk sheet/strip material used for washer fabrication, may also be included with the semi-finished washers prior to heat treatment. If necessary, tensile testing can be performed by the vendor to resolve discrepancies in the hardness test results. Variation in the readings shall not exceed 4 HRC points. No hardness reading shall be less than HRC 42 or greater than HRC 50. Testing will be considered acceptable if temperature sensors indicate conformance with SAE-AMS-5520 conditioning and precipitation heat treatment temperatures and holding times are within the required tolerances, and if the above hardness requirements are met. Hardness readings resulting from acceptable testing shall be averaged to create a typical hardness value for use as a standard (±2 HRC points) for production hardness tests.

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4.4.1.8.2 Compression testing. The following tests shall be done on selected bolt assemblies or individual washers as indicated:

a. Load-deflection test (bolt assembly). It must be verified that washers are in proper parallel-series sequence. Collars need not be installed. Each sample shall be compressed on a tensile test machine or equivalent rigid test frame calibrated for the accuracy required. Results must be adjusted to eliminate deflection of the test frame or machine compliance of the tensile machine. The bolt specimen shall be supported in a hollow cylinder that fully supports the washer while permitting compression of the washers. The test fixture cannot interfere with the portion of the bolt that extends below the washer stack. Curves of compression load (in pounds) versus deflection of the three bolt assembly washer stacks over the range of 0 to 0.040 inch deflection shall be generated.

b. Permanent set. Three individual washers shall be deflected to 0.140 inch overall height (⅔ of the formed height). Before and after the test the overall height shall be measured.

c. Repeated loading. Three individual washers shall be deflected 10 times from free height to within 0.004 to 0.006 of solid height (flat) and visually inspected.

4.4.2 General examination. The gauge assemblies with valves, if applicable, shall be subjected to a thorough examination to determine that the material, workmanship, safety to operating and maintenance personnel, design, and construction are in conformance with this specification and applicable drawings. Examination shall include, but not be limited to, the following:

a. Verification that the arrangement of body sections, nipples, valves, insert recesses, chamber passages, and covers and cover clamping devices meet specific requirements for the gauge type. Verification that the required visibility is provided for the gauge type.

b. Verification that fasteners and locking devices are of the correct type, and that thread protrusion beyond nuts meets requirements. Inspection for coating and plating that may be either required or prohibited on fasteners. Inspection for tack welds required by 3.6.

c. Verification that gasket seating surfaces on gauges, flanges (including nipples), and valves are suitable for the specified gasket materials based on surface finish requirements herein. This inspection will also include valve packing gland design and suitability for the required packing materials.

d. Verification of the compatibility and fit of closed illuminators or other attached light assemblies and valve sheave wheels with chain guides. Verification that illumination devices have the same hardware arrangement as tested under 4.5.1 and fit securely to gauge body.

4.5 Sampling for conformance inspection.

4.5.1 Lot. All items of the same type, configuration class, length, composition, size, and pressure rating (see 1.2) presented for delivery at one time shall be considered a lot.

4.5.2 Sampling for visual and dimensional examination. A random sample of items, except Type A shut-off valve assemblies, shall be selected from each lot for examination of visual and dimensional characteristics in accordance with ASQ H1331 at Inspection Level III. These sample items shall be examined as specified in 4.6. Sample size for valve assemblies is shown in table VI.

TABLE VI. Valve sample size.

Lot Size	Sample Size
2 to 25	1
26 to 65	2
66 to 180	3

4.5.3 Sampling for tests. A random sample of items shall be selected from each lot in accordance with ASQ H1331 at Inspection Level III for the tests specified in 4.7.

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4.6 Conformance inspection.

4.6.1 Visual and dimension examination. Each of the sample items selected as specified in 4.5.2 shall be visually and dimensionally examined to verify compliance with all requirements of this specification not involving tests, and for conformance to 5000-S5100-841604.

4.6.2 Magnetic particle inspection (type A gauges only). Pressure containing alloy forgings shall be 100 percent magnetic particle inspected in accordance with T9074-AS-GIB-010/271. Carbon steel forging are not to be magnetic particle inspected.

4.7 Tests. Each of the samples selected as specified in 4.5.3 shall be subjected to the applicable tests listed in table VII. When specified (see 6.2), tests specified in table VII may use an AQL of 1.0.

TABLE VII. Tests.

Test	Reference Paragraph
Hydrostatic pressure, valve shell	4.7.1.2
Hydrostatic pressure, assembled valve (for seat tightness)	4.7.1.3
Hydrostatic pressure, ball-check	4.7.2
Spring cone hardness	4.7.6.2
Type A glass sets, first article and conformance	4.7.7
Bend test (glass)	4.7.7.7
Polarization test (glass)	4.7.7.8
Mica visual quality (Type A only)	4.7.7.10
Type B and C glass sets, conformance	4.7.8
Gauge Assembly Pressure Tests:	
Hydrostatic pressure, or Steam pressure	4.7.3 4.7.4

4.7.1 Shut-off and drain valve hydrostatic pressure tests.

4.7.1.1 Hydrostatic tests. Each of the sample valves selected as specified in 4.5.2 shall be subjected to the tests specified in 4.7.1.2 for strength and porosity and the tests specified in 4.7.1.3 for tightness. The water temperature shall be 70 to 120 ºF. The test pressure tolerance shall be +2 percent to -0 percent of nominal and rounded off to the nearest multiple of the smallest graduation of the test pressure gauge, not to exceed 50 psig.

4.7.1.2 Shell test. Valve bodies shall be subjected to the applicable pressures specified in table VIII for a duration of 3 minutes. Any weeping, leakage, or permanent deformation shall be cause for rejection.

TABLE VIII. Test pressure for strength and porosity.

ANSI Valve Ratings (psi)	Pressures (psig)
150 (Type C)	450
300 (Type B)	1125
600 (Type A, Class I and II)	2250
1500 (Type A, Class IV)	5575

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4.7.1.3 Seat tests (tightness). Valves shall be pressurized in the direction tending to open the valve and examined for seat tightness. Test pressures shall be as specified in table IX. Stems shall be torqued to 17-23 foot-pounds. Duration of test shall be up to 15 minutes if leakage rate must be measured. There shall be no visible signs of leakage. Should there be any visible leakage, this test shall be continued for 15 minutes to accurately determine the rate of leakage. In checking leakage, three consistent measurements shall be made and recorded for each test. Leakage shall not exceed the amount specified in 3.10.2.2. The maximum allowable force to seat valves during this test shall be as specified in 3.10.2 and 3.10.3. When valve is depressurized, gland and packing shall be removed. The stem shall be backseated to 17-23 foot-pounds of torque. The valve shall be positioned so that leakage can flow by gravity through the stuffing box. The same test pressure durations and leakage criteria as the main seat test shall apply to the back seat test.

TABLE IX. Test pressures for seat tightness.

ANSI Valve Ratings (psi)	Pressures (psig)
150 (Type C)	313
300 (Type B)	816
600 (Type A, Class I and II)	1629
1500 (Type A, Class IV)	4074

4.7.2 Hydrostatic test of ball-check. Each automatic ball-check valve selected as specified in 4.5.3 shall be tested for leakage by hydrostatic pressure. Automatic ball-check valves shall seat automatically under the action of escaping water, shall withstand a hydrostatic pressure of one and one-half times the designated pressure with leakage of not more than 100 cm^3 an hour, and shall open automatically upon release of pressure.

4.7.3 Gauge hydrostatic pressure test. Each gauge shall be subjected for at least 1 minute, maximum 10 minutes, with all valves except the drain valve open, to a hydrostatic pressure equal to the applicable design pressure as a test of the strength and rigidity of the gauge for the pressure at which it is to be used. Glass and gaskets used for this test and 4.7.4 steam pressure test below may be in accordance with the gauge manufacturer's drawings and specifications for the gauge. Any leakage or distortion shall be considered cause for rejection of the gauge being tested.

4.7.4 Gauge steam pressure. Each gauge shall be subjected for at least 1 minute, maximum 10 minutes, with all valves except the drain valve open, to the applicable design pressure. Any leakage or distortion shall be cause for rejection of the gauge being tested.

4.7.5 Valve stem hardness tests. The manufacturer shall perform a Rockwell hardness test on all finished stems after heat treatment. Testing shall be limited to the sheave wheel end of the stem, the stem square, and the surface defined by the packing diameter, within ½ inch of the Acme threads. The stem hardness shall consist of three hardness readings taken at any combination of locations specified above. The hardness criterion of 803-2177525 shall be used to determine acceptability of stem hardness.

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4.7.6 Class IV gauge spring cone washer tests.

4.7.6.1 Compression testing. Complete capscrew (bolt) assemblies with assembled spring cone washers in accordance with 02131 shall be compressed on a tensile test machine or equivalent rigid test frame with hydraulic actuator. Results must be adjusted to eliminate deflection of the test frame or machine compliance of the tensile machine. The bolt specimen shall be supported in a hollow coupling so as to permit compression of the washers without interference from the capscrew portion extending below the washers. A curve of compression load (in pounds) versus deflection of the washer stack over the range of 0 to 0.040 inch deflection shall be generated. The stiffness (slope of curve) in the linear region between 0.010 and 0.020 inches of deflection shall be between 110,000 and 225,000 pounds per inch. Variation between samples tested should not exceed 25,000 pounds per inch.

4.7.6.2 Hardness testing of spring cone washers. Rockwell hardness testing shall be conducted on spring cone washers. A random sample of washers shall be selected from each lot for examination of visual and dimensional characteristics in accordance with ASQ H1331 at Inspection Level III. Testing shall be done at two locations 180 degrees apart on the internal conical surface, midway between the hole and outer edge. Hardness shall be in accordance with standard value established during first article test heat treatment testing.

4.7.7 Type A glass sets.

4.7.7.1 First article inspection. The first article sample shall be subjected to the first article inspection shown in table X.

TABLE X. First article and conformance inspection.

Examination and Test	Requirement	Test	First Article	Conformance
General examination		4.4.2	X	X
Visual and dimensional examination	3.12.3.1	4.6.1	X	X
Thermal shock	3.12.3.2	4.7.7.5	X	X
Fracture	3.12.3.2.1	4.7.7.6	X	
Simulative service	3.12.3.3 and 3.12.5.4	4.7.7.9	X	X
Bend test [1/]	3.12.3.3	4.7.7.7	X	
Polarscopic examination [1/]	3.12.3.3	4.7.7.8		X
Mica quality	3.12.4.1	4.7.7.10		
Spring cone washer heat treatment	3.12.7	4.4.1.8.a	X	
Capscrew spring cone compression	3.12.7	4.4.1.8.b	X	
Spring cone washer hardness	3.12.7	4.7.6.2		X
Valve stem hardness	3.12.9.2.4	4.7.5	X	X

NOTE:

[1/] These tests are only required when simulative service test requirements of 3.12.3.3 and 3.12.5.4 are not accomplished.

4.7.7.2 Glass set lot. For purposes of sampling, a lot shall consist of all glass sets of one size and type produced in one plant under the same conditions and offered for delivery at one time. A lot shall not exceed 1000 sets.

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4.7.7.3 Glass set sampling. A random sample of glass sets (glasses, gaskets, spacers, and mica shields, as applicable) shall be selected in accordance with ASQ H1331 at Inspection Level III (S-3 for thermal shock, fracture, and service test). Each of the sample sets or units shall be subjected to the conformance inspection shown in table X. When specified (see 6.2), an AQL of 1.5 may be used for Class A glass sets.

4.7.7.4 Visual and dimensional examination. Each of the sample glasses, gaskets, spacers, and mica shields selected as specified in 4.5.2 shall be visually and dimensionally examined to verify compliance with the requirements of this specification not involving tests. Any items in the sample containing one or more visual and dimensional defects shall be rejected, and likewise, the lot represented by the sample shall be rejected.

4.7.7.5 Thermal shock test. The glasses shall be placed on edge in a wire rack which has been wrapped with insulation to prevent glass-to-metal contact. The glasses shall be placed in the rack with care taken to ensure there is no glass-to-glass contact of the samples. The glasses shall be placed in the tack as specified herein and heated in a furnace (preheated to 500 °F) for a period of 25 minutes. The glasses shall then be quickly immersed in a cold water bath maintained at a temperature of 40±5 °F. The glasses shall then be examined to determine conformance to 3.12.3.2.

4.7.7.6 Fracture test. Glasses shall be mounted in a water gauge frame and impacted with a pointed tool, such as a center punch, at the midpoint of the outer face to produce fracture. The glass shall be examined to determine conformance to 3.12.3.2.1.

4.7.7.7 Bend test. The bend test shall be in accordance with DIN 7081 requirements using the sampling requirements of 4.7.7.3. The bend test shall only be accomplished when simulative service testing in accordance with 4.7.7.9 is not required.

4.7.7.8 Polariscopic examination. A polariscope such as specified in ASTM F218 shall be used to examine the glasses sampled by 4.7.7.3 qualitatively for the rings or strain pattern specified in ASTM F218. Each sample shall show clear isochromatic lines running parallel to the edges.

4.7.7.9 Simulative service test for glass set. Type A glass sets (glass, mica, and gaskets) shall be installed in the applicable water level gauge frame which shall be mounted on a boiler steam drum or test facility (simulating boiler conditions) and operated at a steam gauge pressure of approximately 1500 psig. Class IV glass sets shall be hydrostatically tested to 2500 psig. Gauge assembly shall be in accordance with instructions of the gauge manufacturer. Minimum requirements of simulative service test shall be as follows:

a. Assembled water level gauges shall be operated continuously for a minimum of 100 hours at design pressure.

b. Water level shall be maintained at the visible range centerline.

c. Water level gauge shall be blown down a minimum of five times during each working day in accordance with the following procedure:

(1) Open water level gauge drain line stop valve.

(2) Shut water level gauge top cut out valve.

(3) Crack open water level gauge drain valve and blow down gauge, clearing it of all water.

(4) Shut water level gauge drain valve.

(5) Open water level gauge top cut out valve.

(6) Shut water level gauge bottom cut out valve.

(7) Crack open water level gauge drain valve and blow down gauge for 20 seconds.

(8) Shut water level gauge drain valve.

(9) Open water level gauge bottom cut out valve. Ensure water level returns to visible range centerline.

(10) Shut water level gauge drain line stop valve.

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4.7.7.10 Mica quality test. Visual quality tests in accordance with ASTM D351 shall be applied to each lot, checking carefully for flaws and surface defects. Air bubbles (inclusions) and waviness rather than stains (discoloration) shall determine mica quality, unless the stains are too dark for clear visibility in the usable (vision) area of the mica shield. Samples shall be measured for compliance with minimum thickness standards of 3.12.4.2 or 3.12.4.3. Samples shall be checked for compliance with the film, visible scratch, edge fracture, and manufacturing standards specified in 3.12.4.4.

4.7.7.11 Tests for quality of sheet gasket material for classes I and II. No additional tests are required for MIL-DTL-24696 gasket and spacer material.

4.7.8 Type B and C glass sets. Conformance inspection shall be performed on the sample selected as specified in 4.7.8.1. This inspection shall include examination of 4.7.8.2 and the tests of 4.7.8.3.

4.7.8.1 Lot and sampling. For the purpose of sampling, a lot shall consist of all glass sets of one size and type produced in one plant under essentially the same conditions and offered for delivery at one time. A lot shall not exceed 1000 sets. A random sample of glass sets (glass and gaskets) shall be selected in accordance with ASQ H1331 at Inspection Level S-2. When specified (see 6.2), AQL of 2.5 may be used for Class B and C glass sets.

4.7.8.2 Examination. Each sample selected as specified in 4.7.8.1 shall be examined for compliance with the requirements of this specification. This inspection shall encompass all visual examinations and dimensional measurements.

4.7.8.3 Tests. Each sample selected in accordance with 4.7.8.1 shall be tested to determine conformance to the requirements of this specification. Tests shall be conducted as specified in 4.7.8.3.1 and 4.7.8.3.2.

4.7.8.3.1 Corrosion test. The corrosion test shall be performed as follows:

a. Glass shall be carefully placed in gauges with proper gaskets in the front and back of the glass.

b. Dry saturated steam shall then be allowed to pass through the gauge for a period of 48 hours at a pressure of 275 psig.

c. Glass shall show no surface disintegration.

4.7.8.3.2 Spall test. Glasses shall not spall or crack when dropped flatwise into cold water at 48±5 °F after heating for 15 minutes at 350±5 °F.

4.8 Toxicity. A Health Hazard Assessment shall be conducted to ensure conformance to 3.19, as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the material using the administrative Health Hazard Assessment (HHA) data provided by the manufacturer/distributor to the NMCPHC.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type A and B gauges including replacement glass sets are intended for direct boiler-water reading on US Naval surface ships. Type C indicators, including gaskets, glass, and glass sets are intended for boiler applications not covered by Types A and B, and direct reading tank level applications on U.S. Naval ships. The replacement gaskets alternatives for Type C gauges are intended to allow specific materials shown on detail drawings to be procured under this document. This document should not be used where specific gasket shape/size and material requirements are available on separate NAVSEA drawings such as 820-5348665 for nuclear plant applications.

6.1.1 Type A. Type A gauges are for use on main boilers and large auxiliary boilers.

6.1.2 Type B. Type B reflex gauges are for use on auxiliary boilers.

6.1.3 Type C. Type C gauges are for use on waste heat boilers requiring 150 psig flanged reflex gauges. These gauges and welding pad designs included in Type C are used in a variety of tank-level indicating applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. When first article inspection is required (see 3.1).
d. When bolts, studs, nuts, and screws are other than specified (see 3.6).
e. Chain requirements for sheave wheel or lever pull valves if other than as required in 3.10.2.1.
f. When valve packing should be removed prior to shipment (3.10.5).
g. When valve packing should be shipped separately (3.10.5).
h. When Class B winding material is a requirement for MIL-G-24716 gaskets (3.10.5).
i. Quantity and size of drain valves required (see 3.10.6).
j. Visibility range required for Type A, if other than specified (see 3.12.2).
k. Shape and dimensions for Type A, if other than specified (see 3.12.3.1).
l. When special 22-degree bevel glass is required for size 7 glass (3.12.3.4).
m. When mica shield is not required (see 3.12.4.2).
n. When two piece mica shield is required (3.12.4.3).
o. When other than two gaskets are required for Class I gauges (see 3.12.5.2).
p. Pressure rating of mounting flanges for Type A gauges, if other than specified (see 3.12.9.4).
q. Size and visibility range of reflex glass for Type B and C gauges (see 3.13.1).
r. Quantity, type, and size of drain valves required for Type B and C gauges (3.13.4, 3.13.8, and 3.14.7).
s. When shock test is required for Type B and C gauges (see 3.13.6 and 3.14.8).
t. When vibration testing is required for Type B and C gauges (see 3.13.7 and 3.14.9).
u. When specific material is identified for Type C welding pad gauges, and when welding insert bar is to be provided (see 3.14.2).
v. When special cleanliness and detrimental material requirements apply to glass, gaskets and mica (3.14.5, 3.17, and 3.18).
w. When level indicating plates are required (see 3.15.1).
x. When special cleanliness is required for gauges and valves (3.17).
y. Requirements for a Health Hazard Assessment (see 3.19 and 4.8).
z. When vibration testing is required for Type B and C gauges (see 4.4.1.2).

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aa. When the Acceptance Quality levels (AQL) of 2.5 can be used for Class B and C glass sets (see 4.7.8.1), when an AQL of 1.5 can be used for Class A glass sets (see 4.7.7.3), and when tests can use an AQL of 1.0 (see 4.7).

bb. Packaging requirements (see 5.1).

cc. When certificates of compliance should be provided (see 6.4).

dd. Levels of preservation and packing required (see 6.8.2, 6.8.3, and 6.8.3.1).

ee. When waterproof caseliners are not required (see 6.8.3.1.1.1).

ff. Special marking required (see 6.8.4.1).

6.3 Applicable documents.

6.3.1 General. The documents listed in this section are specified in section 6 only of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples.

6.3.2 Government documents.

6.3.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated

MIL-PRF-23199 - Packaging and Packing Requirements for Special Purpose Mechanical Components and Repairs Parts

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-767 (Classified) - Control of Hardware Cleanliness

MIL-STD-2073-1 - Standard Practice for Military Packaging

(Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil.)

(Copies of MIL-STD-767 may be requested from commandstandards@navy.mil addressed with a need-to-know.)

6.3.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D3951 - Packaging, Commercial

ASTM-D4727 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

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6.4 Certificates of compliance. Certificates of compliance should be provided for all tests required in section 3 (see 6.2). The certificates should include found test results, the specified requirements, and the conclusion that each test passed. Certificates of compliance should also be required for FAT certification of heat treatment of spring cones and compression testing of spring cone bolt assemblies. This certification should include the following data: quantity heat treated (lot), and for each lot the actual temperature range and duration of each heat treatment cycle. Heat treat furnace charts which show the date/time that the FAT lot was moved in/out of the furnace should also be included. The certification should validate that the washers have not been exposed to chlorides at any point in the procedure. Plotted curves generated by first article compression testing of bolt assemblies should be included with compliance data.

Certificates should be provided to certify that gasket and packing materials be in accordance with military specifications, manufacturer's specifications or ASTM F104 line call outs. A copy of each certificate should be provided to Commanding Officer, Naval Surface Warfare Center, Carderock Division, Ship Systems Engineering Station, Attn: NSWCCD-SSES Code 922, 5001 South Broad St., Philadelphia PA 19112 in addition to other reporting requirements of the contract.

6.5 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of one unit of each class. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Provisioning. Provisioning Technical Documentation (PTD) spare parts and repair parts should be furnished as specified in the contract.

6.6.1 Spare parts. When ordering spare parts or repair parts for the equipment covered by this specification, the contact should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of equipment. Insert parts (glass, gaskets, and mica) offered by gauge manufacturers, and included in original equipment supplied under this document, may not be identical to spare glass set items procured to the requirements of this document. Packaging for such parts should also be specified.

6.7 Sub-contracted material and parts. The packaging requirements of the referenced documents listed in 6.3 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Preservation, packing and marking instructions.

6.8.1 General.

6.8.1.1 Navy fire-retardant provisions.

6.8.1.1.1 Lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing should be fire-retardant treated material in accordance with MIL-L-19140 as follows:

Level A and B	Type II resistant Category I – general use
Level C	Type I – non-weather resistant Category I – general use

6.8.1.1.2 Fiberboards. Fiberboards used in the construction of class-domestic non-weather resistant fiberboards and cleated fiberboards boxes should meet the flame spread index and the specific optic density requirements as specified in ASTM D4727 and amendments thereto.

6.8.2 Preservation. Preservation should be level A, C, or commercial as specified (see 6.2).

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6.8.2.1 Level A.

6.8.2.1.1 Cleaning and drying. Cleaning and drying of all gauge (see 3.17) surfaces (internally and externally), glass sets, and parts should be by a process and procedure in accordance with MIL-STD-2073-1 which should ensure removal of corrosion, dirt, grease, and other foreign material without damage to the item.

6.8.2.1.2 Gauges.

6.8.2.1.2.1 Preservation. Application to chamber inner surfaces of contact type preservation compounds should not be required. Molybdenum-disulfide coating should be applied to frame bolts or studs. Frame insert gasket seating surfaces may be similarly protected when glass sets are not installed for shipment. Manufacturer's commercial practice may also be followed if preservative used complies with other requirements.

6.8.2.1.2.2 Unit protection.

6.8.2.1.2.2.1 Gauges weighing less than 40 pounds. Gauge openings should be sealed with water resistant, pressure sensitive tape. Gauges should be individually packed in water-resistant folding, set-up, or metal edge paperboard or fiberboard boxes in accordance with the unit container requirements of MIL-STD-2073-1 at the contractor's option. Box closure should be in accordance with the applicable box specification. Gauges should be supported, blocked, braced, or cushioned to prevent damage during handling, shipment, stowage, and storage.

6.8.2.1.2.2.2 Gauges weighing 40 pounds and over. Gauges should be disassembled and each frame assembly and each valve individually packed in boxes (see 6.8.2.1.2.1). Detached connecting hardware should be placed in waterproof envelopes or bags, sealed, and included in the component box. The boxed items, making up a complete gauge, should then be packed together as a unit in containers for the level of packing specified.

6.8.2.1.3 Glass sets. One glass, two gaskets (on inner and one outer gasket), two spacers (as applicable), one mica, and one mica shield (consisting of one or two pieces) should be packed in boxes (see 6.8.2.1.2.2.1). Mica shields should be placed in a plastic or kraft paper envelop or bag, reinforced by heavy cardboard on each side of the mica, and inserted in the carton or box in a position between the gaskets and glass. For glass traceability, each glass shipping carton containing more than one individual glass should contain a production lot identification number to which the manufacturer has provided certifications for the fracture and thermal shock testing requirements (bend test and polariscopic examination requirements, when applicable). Glass shipped from the manufacturer should be shipped in containers that contain glass from a single lot only. This provision should not be applicable if the individual glass boxes display lot identification numbers.

6.8.2.1.3.1 Gaskets. When ordered separately, two gaskets (one inner and one outer gasket) should be unit packed in a kraft paper envelope and six envelopes (12 gaskets) should be packed in a unit container specified in 6.8.2.1.2.2.1. Envelopes should contain a fiberboard insert that will prevent gasket damage from bending.

6.8.2.1.3.2 Mica shields. When ordered separately, one mica shield (consisting of one or two pieces) should be unit packaged in a kraft paper envelope or plastic bag, reinforced by heavy cardboard on each side of the mica. When required, the thicker of the two piece mica shields should be identified on the packaging. Twelve envelopes or bags should be packed in a unit container specified in 6.8.2.1.2.2.1.

6.8.2.1.4 Valves. Valve preservation and packing should be in accordance with 803-2177525. Marking on individual graphitic seal ring packaging should include packing dimensions; Inner Diameter (ID) by Outer Diameter (OD) by height.

6.8.2.1.5 Spring cone washers. Spare spring cone washers, and bolt assemblies with captive spring cone washers, should be sealed in plastic bags selected to protect against exposure to marine environment while stored aboard a ship. Bolt assemblies with captive spring cone washers should be individually packaged. Individual spring cones should be packaged in quantities of three or to match number required at each stud of the water gauge.

6.8.2.2 Level C. Preservation of gauges, glass sets, gaskets, and mica shields should be as specified for Level A except that the unit containers specified in 6.8.2.1.2.2.1 should be of the domestic or nonweather resistant type, class, or variety as applicable.

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6.8.2.3 Commercial. Commercial packaging (cleaning, preservation, cushioning, and the unit pack) should be in accordance with ASTM D 3951. Unit pack quantities should be as specified under level A.

6.8.3 Packing. Packing should be level A, B, C or commercial as specified (see 6.2). Where cleanliness has been maintained in accordance with MIL-STD-767, the requirements of MIL-PRF-23199 packaging should be specified (see 6.2).

6.8.3.1 Level A, B, and C containers. Materials preserved as specified (see 6.8.2) should be packed in shipping containers, for the level of packing specified (see 6.2), in accordance with the exterior shipping container tables of MIL-STD-2073-1 and herein. Unless otherwise specified (see 6.8.2.1), container selection should be at the contractor's option. Glass sets should be packing in units of one dozen or multiples thereof. Packages containing glass should be separated by fireboard pads, and lined top, bottom, and sides with double wall corrugated fiberboard.

6.8.3.1.1 Caseliners, closure, and gross weight.

6.8.3.1.1.1 Caseliners. Unless otherwise specified (see 6.2), Level A shipping containers should be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

6.8.3.1.1.2 Closure. Container closure, reinforcing, or banding should be in accordance with the applicable container specification or appendix thereto except that fiberboard boxes should be closed in accordance with 2B2 or 4B5of ASTM D1974/D1974M, as applicable. Weather resistant fiberboard boxes should be reinforced with non-metallic or tape banding.

6.8.3.1.1.3 Weight. Wood, plywood, and cleated-type containers exceeding 200 pounds gross weight should be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

6.8.3.2 Commercial. Material preserved as specified (see 5.1) should be packed for shipment in accordance with ASTM D3951 and herein.

6.8.3.2.1 Container modification. Shipping containers exceeding 200 pounds gross weight should be provided with a minimum of two, 3- by 4-inch nominal wood skid laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, storage, and stowage.

6.8.4 Marking.

6.8.4.1 Levels A, B, C, and commercial. In addition to any special marking required (see 6.2), interior packs and shipping containers should be marked, including barcoding, in accordance with MIL-STD-2073-1.

6.9 Subject term (key word) listing.

Borosilicate
Fittings
Indicator
Liquid level
Mica shield
Reflex
Visibility
Water gauge
Welding pad

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6.10 Definitions.

6.10.1 Special tools. Special tools are defined as those tools not listed in the Federal Supply Catalog [copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)].

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 6680-2012-001)

Review activities:
Navy – MC, YD
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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